

IMAGINE WAKING up tomorrow to a world that has suddenly fallen silent. You reach for your smartphone, but the screen stays black. You walk out to your car, but the engine won't start. At the local store, your UPI payment fails, and down the street, hospital monitors go dark. This is not a scene from a dystopian movie but a glimpse of what happens when the supply of a single component dries up.

Silicon chips are the invisible nervous system of modern economies, powering everything from digital payments and telecommunications networks to automobiles, medical devices and critical infrastructure. Yet, as India races towards becoming a digital superpower, it faces a fragile paradox. The brains powering much of the world's technology may be designed by Indian engineers, but the chips themselves carry foreign passports. Largely, their silicon is birthed in Taiwan, stamped with American patents, and packaged somewhere in Southeast Asia.

In an era where technology has become a strategic asset, disruptions caused by geopolitical tensions, trade restrictions, natural disasters or supply-chain shocks can ripple through entire industries. This leaves India's digital economy exposed to forces beyond its control.

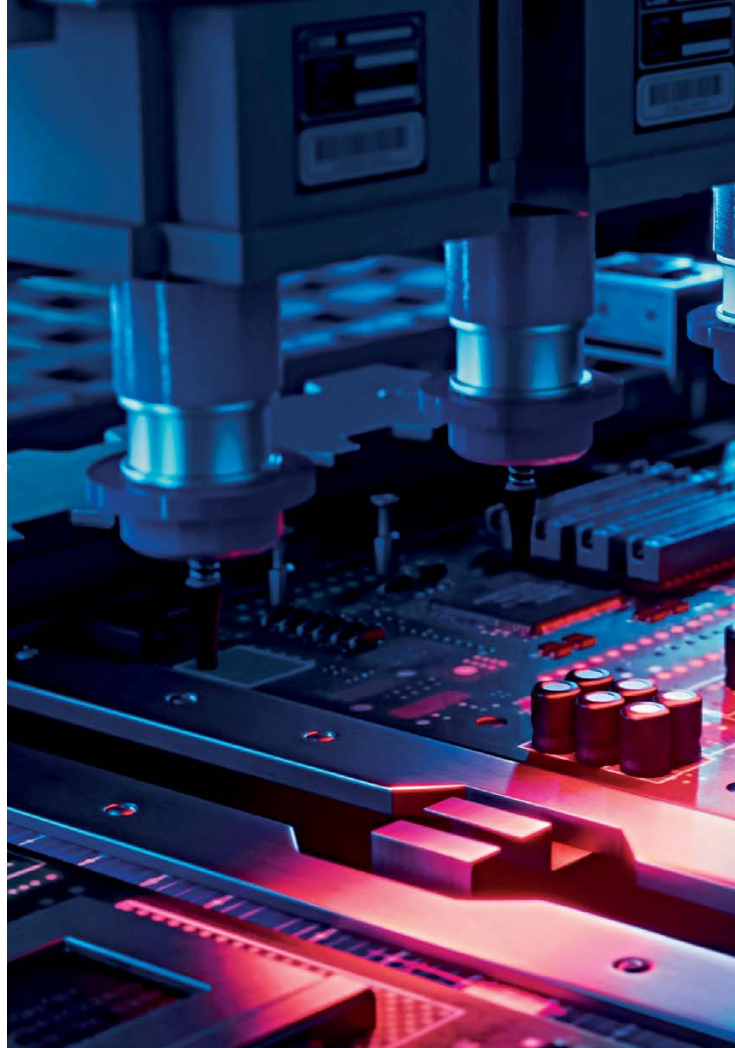
50 | THE SILICON DEFICIT

The scale of India's semiconductor dependence becomes clear when measured in dollars. India's semiconductor consumption in 2025 was estimated at \$30-35 billion, with the overwhelming majority of that demand being met through imports. "Majority of it was import dependent and will continue to be so for the next couple of years, till the local semiconductor manufacturing reaches steady-state yields," says Apaar Bhatnagar, Associate Partner, IM & Auto, KPMG in India.

India's appetite for chips has expanded rapidly alongside the digitisation of the economy. Between FY17 and FY25, India spent nearly \$150 billion importing semiconductor products, with imports growing at a compound annual growth rate of 23%. According to estimates by the Ministry of Commerce and Industry, if current trends persist, the annual semiconductor import bill could swell to nearly \$240 billion by 2035.

The challenge is that demand is only beginning to accelerate. "As the demand for semiconductors grows, volumes are expected to be driven by AI data centres (25%), consumer electronics (30%), automotive (20%), industrial automation (15%) and telecom infrastructure and other industrial applications (10%)," says P.S. Subramaniam, Partner in the Strategic Operations practice at Kearney.

The concern is not that India lacks access to semi-



conductors today but that so much of that access depends on events unfolding far beyond its borders. When the bulk of a critical technology is sourced from overseas, supply security becomes inseparable from geopolitics, a reality that has pushed semiconductors from the realm of commerce into the domain of national strategy.

Recognising that technological sovereignty cannot rest on permanent dependence in today's world, the central government has committed billions of dollars to build a domestic chip ecosystem spanning chip fabrication, packaging, testing and design under the India Semiconductor Mission.

But given the complexity and global nature of semiconductor supply chains, experts believe imports will remain an integral part of India's technology ecosystem for years to come. "Given the ongoing efforts, India could realistically produce about 50% of its semiconductor demand locally," says Subramaniam.

But breaking free from chip imports is not as simple as building a factory and turning on the lights. The paradox of semiconductor self-reliance is that the road to domestic manufacturing is entirely paved with foreign tools.